

4.4 SUPER CLASS: AGNATHA (Gr. A- WITHOUT; GNATHOS- JAW)

They are jawless primitive fish like aquatic vertebrates without paired appendages. The internal ear bears one or two semicircular canals. Notochord persists in the adult.

Agnatha includes the extant class **Cyclostomata** besides the extinct class **Ostracodermi**.

4.4.1 Class: Cyclostomata (Gr. *cyklos*- circular; *stoma*- mouth)

Cyclostomes are jawless aquatic (marine or freshwater) forms. Body is scaleless, long, slender and eel-like. Endoskeleton is cartilaginous. Vertebrae are represented by imperfect neural arches in some. Mouth is circular and suctional. Tongue bears **horny teeth**. Respiratory organs include six to fifteen pairs of gill slits. Heart is **two-chambered** as in fishes. **Renal portal system is absent**. Kidneys are **mesonephric**.

e.g. *Petromyzon* (lamprey) and *Myxine* (hag fish/slime eel).

Marine **lampreys** migrate to freshwater for spawning (anadromous migration). After spawning they die. Their larvae are **ammocoete** metamorphose into adults and return to the sea.

Hagfishes are exclusively marine. They are considered scavengers because they feed on dead or dying fish. When disturbed hag fish releases a milky fluid from its mucous sacs which becomes slime on contact with sea water. Development is direct.



Fig. 4.6 *Petromyzon*

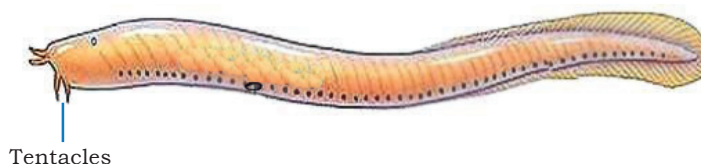


Fig. 4.7 *Myxine*

4.5 SUPER CLASS: GNATHOSTOMATA (Gr. GNATHOS- JAW; STOMA- MOUTH)

They are vertebrates with jaws. They possess paired appendages (pectoral and pelvic fins or fore and hind limbs) and internal ears with three semicircular canals each. Notochord is replaced partly or wholly by the vertebral column.

Gnathostomata includes **jawed fishes** (Pisces) and **Tetrapods** (Amphibia, Reptilia, Aves and Mammalia).

- Pisces and Amphibia constitute the group **Ichthyopsida**
- Reptilia and Aves constitute the group **Sauropsida**.